# Basilisk MPI CPU Benchmarks for Multiphase Flow

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**Abstract**

Basilisk is an open-source framework for solving interfacial flows on quadtree and octree meshes. Its pressure projection places a multigrid Poisson solve inside every incompressible time step, making communication across mesh levels a direct measure of the strong-scaling limit. This report sets out a portable CPU benchmark suite built from two stock Basilisk kernels and a set of two-phase calculations. Measurements on MareNostrum 5 and Snellius cover up to 16,384 MPI ranks for fixed-size kernel problems and up to 1,024 ranks for the Marangoni application cases. Three axisymmetric uniform-grid kernels—bursting bubble, elastic Taylor–Culick retraction and Newtonian drop impact—are timed at $L = 9$, 10 and 11. The $L = 9$ and $L = 10$ series cover 2–768 ranks. $L = 11$ Taylor–Culick and drop impact extend to 2304 ranks so the strong-scaling knee is in the recorded data; bursting $L = 11$ already turns up at 768 ranks. One three-dimensional kernel—viscoelastic drop impact—remains a one-node $L = 7$ sweep. The kernel data support small-to-medium system qualification and expose the onset of communication-limited scaling. Tests above 100 fully populated 192-core nodes remain a proposed campaign rather than a demonstrated result.

## 1 What the benchmark measures

An incompressible Basilisk calculation advances advection, viscosity and interfacial forces before projecting the velocity field onto a divergence-free space. The projection requires a multigrid Poisson solve. Restriction and prolongation transfer information between mesh levels, while the domain decomposition exchanges data between MPI ranks. At sufficiently large rank counts, this communication ceases to be hidden by the work performed on each rank. A fixed-size rank sweep therefore identifies both the useful strong-scaling range and the point at which adding CPUs no longer reduces the time to solution.

The proposed suite has three layers:

1. stock Poisson and Laplacian kernels for reproducible machine-to-machine comparison;
2. two-phase timing workloads that exercise the same solver stack with interface reconstruction and surface tension: the Marangoni series includes a terminal-velocity reference comparison, while the four uniform-grid application kernels provide scaling timings without a separate physical-validation result here; and
3. a workload series in which spatial resolution or the number of drops is increased, testing whether a larger application postpones strong-scaling saturation.

The source cases, compact timing tables and plotting scripts are maintained at `https://github.com/comphy-lab/hpc-basilisk-scaling`. The stock kernel sources are retained without modification from the corresponding Basilisk tests [1, 2].

## 2 Poisson and Laplacian kernel scaling

The two-dimensional kernel uses a full quadtree at level $L = 14$, corresponding to $2^{28}$ cells. Figure 1 compares wall-clock and MPI time on MareNostrum 5 and Snellius with the published Curie measurements. On Snellius, the Poisson time continues to decrease through 8,192 ranks and then plateaus at 16,384 ranks. The simultaneous rise in the communication fraction identifies the large-rank regime as communication limited rather than compute limited.

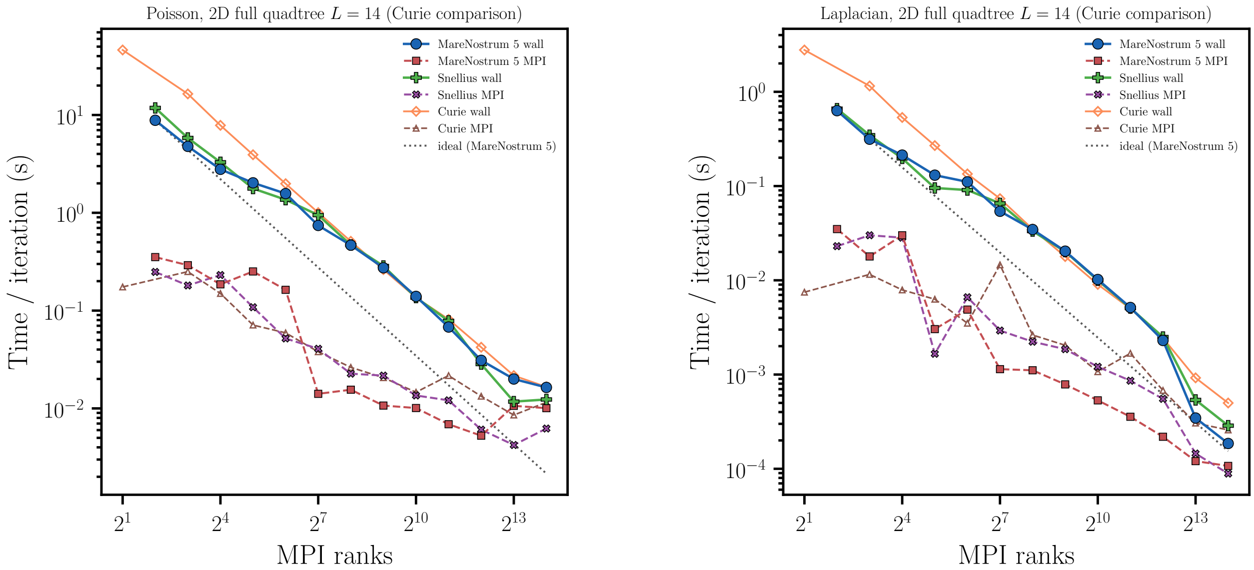


Figure 1: Strong scaling of the Poisson solve (left) and Laplacian operator (right) for the two-dimensional full-quadtree kernel at $L = 14$. Solid curves show wall time per iteration and dashed curves show the measured MPI contribution. The grey dotted line is ideal scaling from the MareNostrum 5 baseline. Curie data are the reference measurements distributed with the Basilisk test.

The three-dimensional kernel uses a full octree at $L = 9$, or $2^{27}$ cells. On Snellius, the Poisson time falls from 19.34 s at two ranks to 0.0333 s at 16,384 ranks. Most of the reduction has already been obtained by 4,096 ranks, where the corresponding time is 0.0390 s. Figure 2 therefore supplies both a broad scaling curve and a clear diagnostic of the strong-scaling limit. The Occigen measurements provide an independent reference for the same stock calculation.

A low-rank kernel timing measures core and memory performance, while the bend in its rank sweep measures the communication stack and collective operations. Both are relevant to production multiphase-flow calculations.

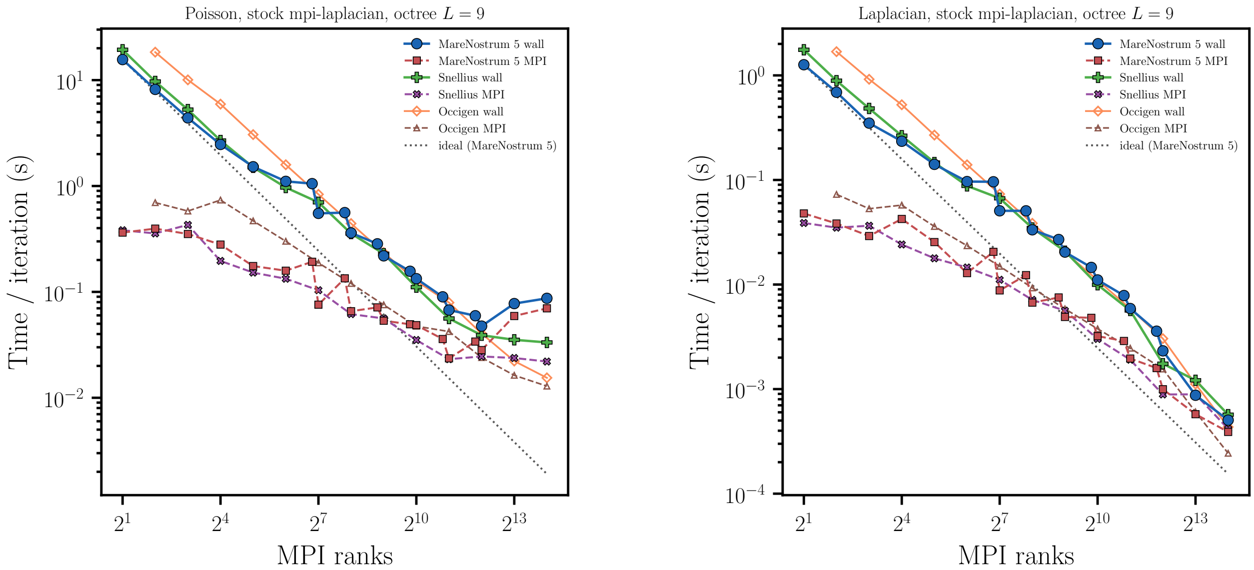


Figure 2: Strong scaling of the stock three-dimensional `mpi-laplacian` test on a full octree at $L = 9$. Symbols show MareNostrum 5 and Snellius measurements alongside the Occigen reference. Solid curves give wall time per iteration and dashed curves give MPI time. The flattening of the Poisson curve beyond a few thousand ranks marks the fixed-problem communication limit.

# 3 Active-drop scaling evidence

The application benchmark follows a drop driven by a surface-tension gradient. The numerical formulation is the conservative, balanced treatment of interfacial stresses described by Abu-Al-Saud, Popinet and Tchelepi [4]. The cluster case adds timing, restart and controlled output to the public Basilisk test without changing the physical validation target.

Figure 3 is the scaling figure used in the final MareNostrum 5 report. It varies the amount of work in two ways. Panel (a) increases the spatial resolution of an axisymmetric uniform mesh. Panel (b) increases the number of drops in a planar domain while retaining 64 points per radius. The larger problems continue to benefit from additional ranks after the smaller problems have saturated. This is the expected application behaviour: the useful rank count grows with the work available per rank. The present Marangoni dataset stops at 1,024 ranks, approximately 5.3 fully populated 192-core nodes, and should be used as small-scale evidence.

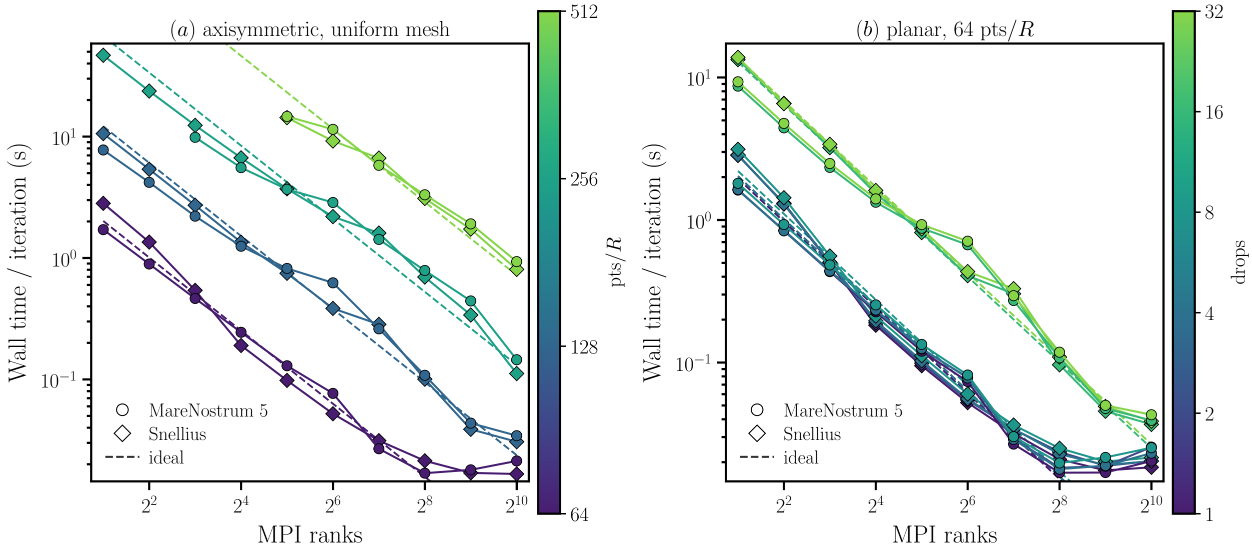


Figure 3: Abundant-fuel active-drop scaling on MareNostrum 5 GPP (circles) and Snellius Genoa (diamonds). (a) Uniform meshes at 64–512 points per radius. (b) Planar 1–32-drop workloads at 64 points per radius. Dashed lines show ideal scaling. Each point is one recorded run; the current tables do not measure run-to-run variability.

## 4 Reference comparison and benchmark coverage

The Marangoni application remains useful only if it solves the intended flow. Figure 4 compares the computed drop speed with the analytical terminal velocity and with the Basilisk reference data. The coarse-resolution error follows the expected second-order trend through 32 points per radius; the finer results approach the theoretical speed but should not be described as a single second-order convergence range because their error plateaus.

The stock kernels extend the measured rank range beyond the representative application. The Marangoni series combines the timing sweep with the reference comparison above; the four uniform-grid application kernels supply additional two-phase timing cases without a separate physical-validation result in this report.

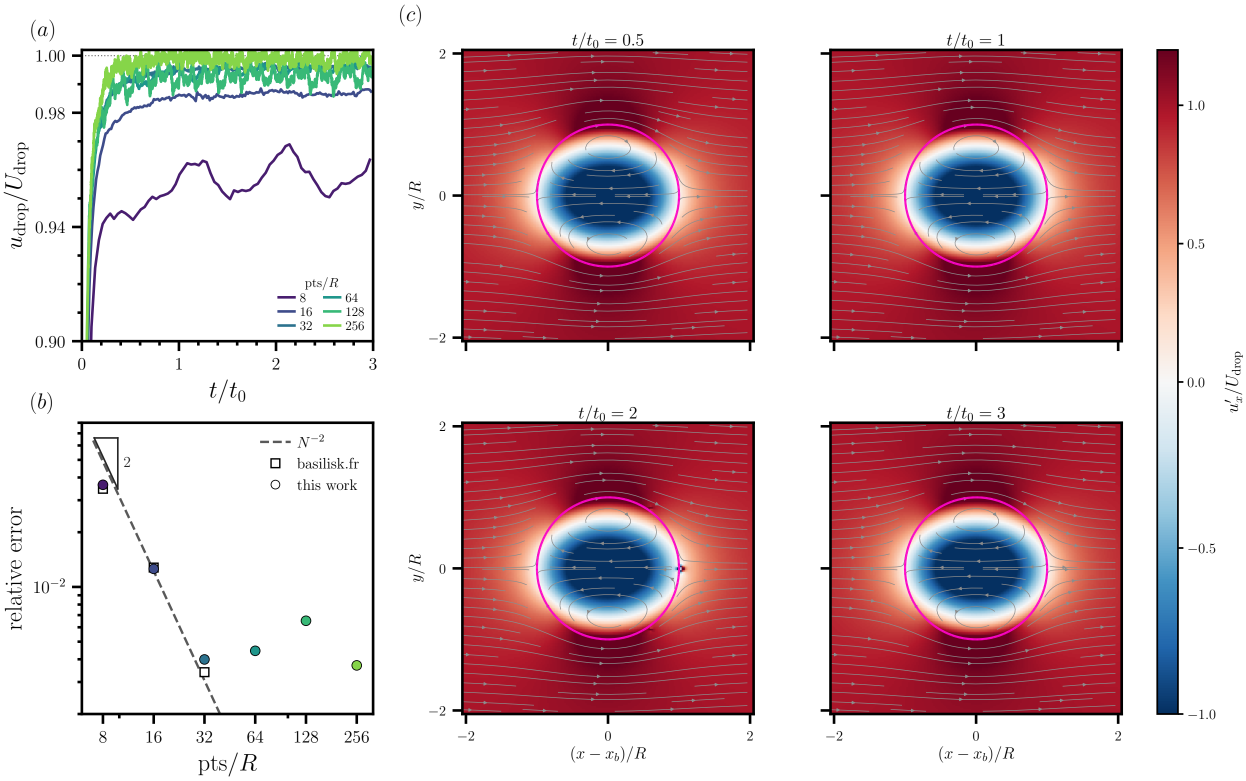


Figure 4: Validation of active-drop migration in the abundant-fuel limit. (a) Speed converges to the Young–Goldstein–Block prediction across 8–256 points per radius. (b) Relative error follows the $N^{-2}$ trend on coarse grids. (c) Drop-frame axial velocity at 64 points per radius; magenta marks the interface.

# 5 Qualification matrix for a 192-core node

The present kernel measurements use powers of two in MPI rank count. They reach 16,384 ranks on Snellius, equivalent to 85.3 fully populated 192-core nodes. This brackets a 64-node test but does not demonstrate the requested class above 100 nodes. Exact node-aligned rank counts must also be tested; performance at 8,192 or 16,384 ranks should not be silently substituted for a 12,288-rank, 64-node measurement.

| Class | Proposed nodes | MPI ranks | Present evidence |
|---|---|---|---|
| Small | 1, 2, 4, 8, 10 | 192–1,920 | Kernel curves bracket the range; Marangoni to 1,024 ranks; axisymmetric Taylor–Culick and drop impact to 2,304 ranks |
| Medium | 16, 32, 64 | 3,072–12,288 | Kernel data at 4,096 and 8,192 ranks; 64 nodes bracketed |
| Large | 128, 256 | 24,576–49,152 | Not yet measured; workload size to be selected after scaling and memory pilots |

Table 1: Proposed node-aligned qualification points for a system with 192 CPU cores per node. Existing evidence and proposed measurements are deliberately separated.

For the small class, the current $L = 9$ three-dimensional and $L = 14$ two-dimensional problems can be retained as fixed-size strong-scaling tests. For 16–64 nodes, the viscoelastic drop-impact (VE3D) case is the lead candidate for a three-dimensional $L = 10$ continuation. Test this case first at 16, 32 and 64 nodes, subject to a memory check. If the $L = 10$ problem leaves too little work per rank at the target scale, a larger problem such as an $L = 11$ octree can then be considered for 128 nodes and, if needed, beyond; a short pilot would establish memory use and initialization cost before that extension. The purpose of the large case is to preserve enough cells per rank to measure the network at scale rather than to time an already saturated problem.

The representative two-phase case should accompany the kernels at small and medium scales. A large-scale application case should be promoted only after its domain size, interface area and output policy have been fixed so that each machine performs the same physical and numerical work.

# 6 Uniform-grid application kernels

The Marangoni series tests one interfacial mechanism while the mesh or the drop count is varied. The four kernels below keep that same uniform-tree MPI stack and change the physics: a bursting bubble, an elastic Taylor–Culick sheet, a three-dimensional viscoelastic impact, and an axisymmetric Newtonian impact. The three axisymmetric kernels are fixed-size strong-scaling tests at $L = 9$, 10 and 11 ($512^2$, $1024^2$ and $2048^2$ cells). $L = 9$ and $L = 10$ cover 2–768 MPI ranks (one to four Snellius Genoa nodes). $L = 11$ Taylor–Culick and drop impact extend to 2304 ranks (twelve nodes) so the recorded series includes the knee; bursting $L = 11$ already turns up at 768 ranks. Colour encodes $N_x = 2^L$, as in Figure 3(a). The three-dimensional kernel remains a one-node sweep at $L = 7$ ($128^3$ cells, 2–192 ranks). The timed window is the incompressible two-phase solver only. Geometric setup is outside that window, as is the serial interface reconstruction and dump used to seed the bursting case. The MPI binaries write no checkpoints. The $L = 11$ axisymmetric series is recorded from eight ranks.

Figure 5(a) is the axisymmetric bursting bubble. At $L = 10$ the wall time per iteration falls from 1.28 s at two ranks to 0.0174 s at 192 ranks and 0.0108 s at 768 ranks. The $L = 9$ curve saturates near 0.008 s beyond 96 ranks. The $L = 11$ curve follows the eight-rank ideal line through 384 ranks (0.0407 s) and rises to 0.0781 s at 768 ranks.

Figure 6(a) is the axisymmetric elastic Taylor–Culick retraction. The log-conformation extra-stress equations ride on the same projection step. At $L = 10$ the wall time falls from 1.85 s at two ranks

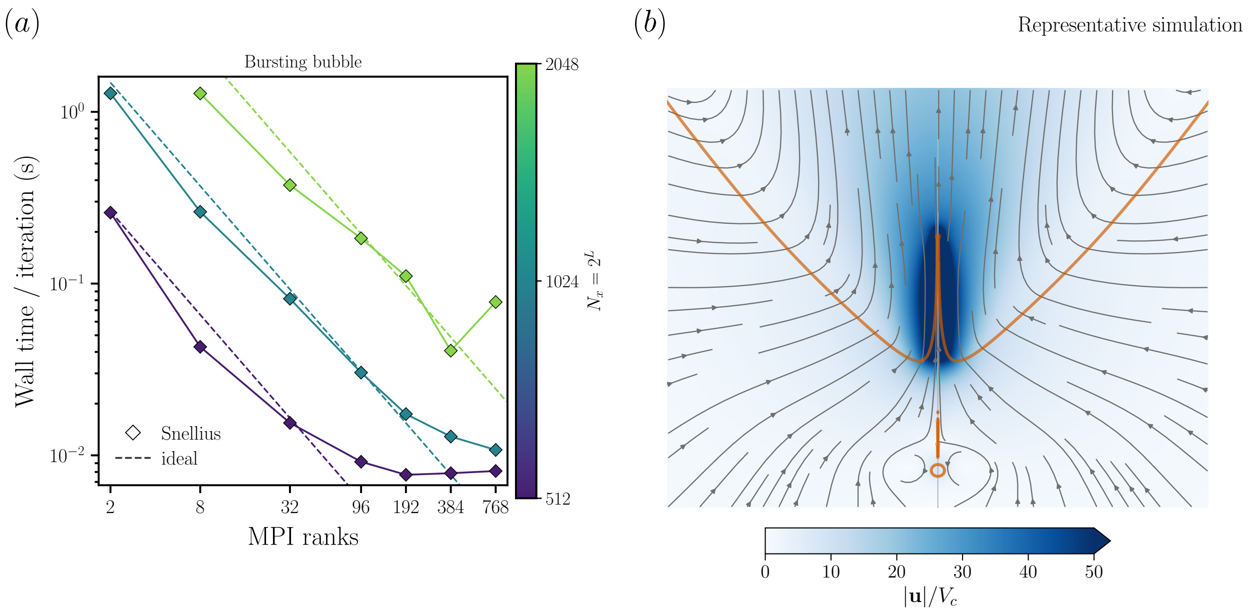


Figure 5: (a) Strong scaling of the uniform-grid bursting-bubble kernel on Snellius Genoa. Axisymmetric Navier–Stokes with VOF and surface tension at $L = 9$, 10 and 11; colour encodes $N_x = 2^L$. Each point is one recorded run of ten solver steps after initialisation; the serial interface reconstruction is not included. The $L = 11$ series starts at eight ranks. (b) Jet and feeding flow from a separate, completed axisymmetric simulation at $Oh = 0.03$ and $Bo = 0$. Blue shows speed normalised by the capillary velocity $V_c = \sqrt{\sigma/(\rho_l R)}$, with colours saturated above $50V_c$; grey curves are instantaneous streamlines and orange marks the interface. The view follows jet inception and includes the small entrapped bubble below the jet.

to 0.0231 s at 192 ranks and 0.0127 s at 768 ranks. The $L = 9$ curve saturates near 0.010 s. The $L = 11$ curve continues to drop, from 1.90 s at eight ranks to 0.0341 s at 768 ranks and 0.0193 s at 1536 ranks, then flattens to 0.0152 s at 2304 ranks.

Figure 7(a) is the axisymmetric Newtonian drop impact on a box of side 8. The drop is placed at $(1.05, 0)$ with impact velocity toward the wall on the left boundary; Basilisk `axi.h` keeps the symmetry axis on the bottom. At $L = 10$ the wall time falls from 2.04 s at two ranks to 0.0267 s at 192 ranks and 0.0173 s at 768 ranks. The $L = 9$ curve saturates near 0.011 s. The $L = 11$ curve follows the eight-rank ideal line to 0.0339 s at 768 ranks and 0.0250 s at 1536 ranks, then rises to 0.0423 s at 2304 ranks.

The three-dimensional kernel uses a uniform octree. Figure 8(a) is a viscoelastic drop impact at $L = 7$ ($128^3$ cells). Wall time per iteration falls from 10.8 s at two ranks to 0.331 s at 192 ranks, a factor of 32. The curve is monotonic but bends away from ideal scaling earlier than the two-dimensional cases. These timings do not separate computation, communication and other overheads, so they do not identify the cause of the bend.

The scaling panels contain Snellius measurements and record a single observation per rank count. The image panels come from separate completed simulations and are not part of the timed trajectories. Compact image inputs, source parameters, attribution and reproduction scripts accompany the figures in the repository.

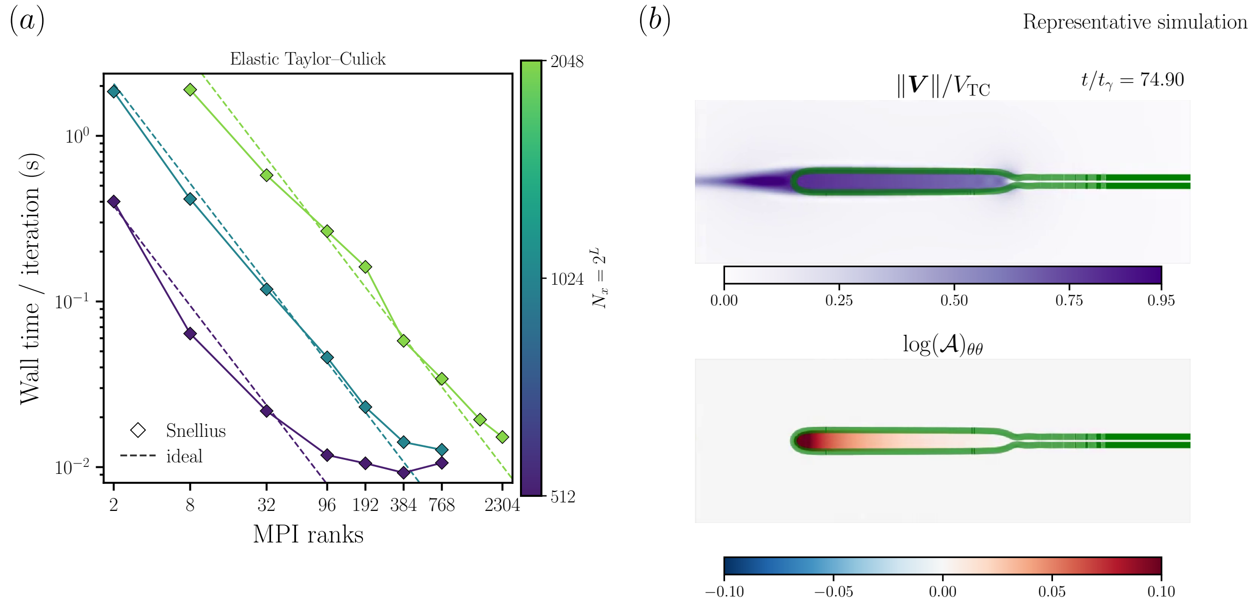


Figure 6: (a) Strong scaling of the uniform-grid elastic Taylor–Culick kernel on Snellius Genoa. Axisymmetric two-phase viscoelastic flow at $L = 9$, 10 and 11; colour encodes $N_x = 2^L$. Timed window: ten solver steps after geometric initialisation of the sheet. The $L = 11$ series starts at eight ranks. (b) Representative axisymmetric elastic retraction at $t/t_\gamma = 74.90$, $Ec = 0.1$ and $Oh = 0.05$. The upper image shows speed normalised by the Taylor–Culick velocity; the lower image shows the azimuthal component $(\log \mathcal{A})_{\theta\theta}$ of the log-conformation tensor. Green marks the interface. Both images show the same rim window from a completed relaxation-free calculation.

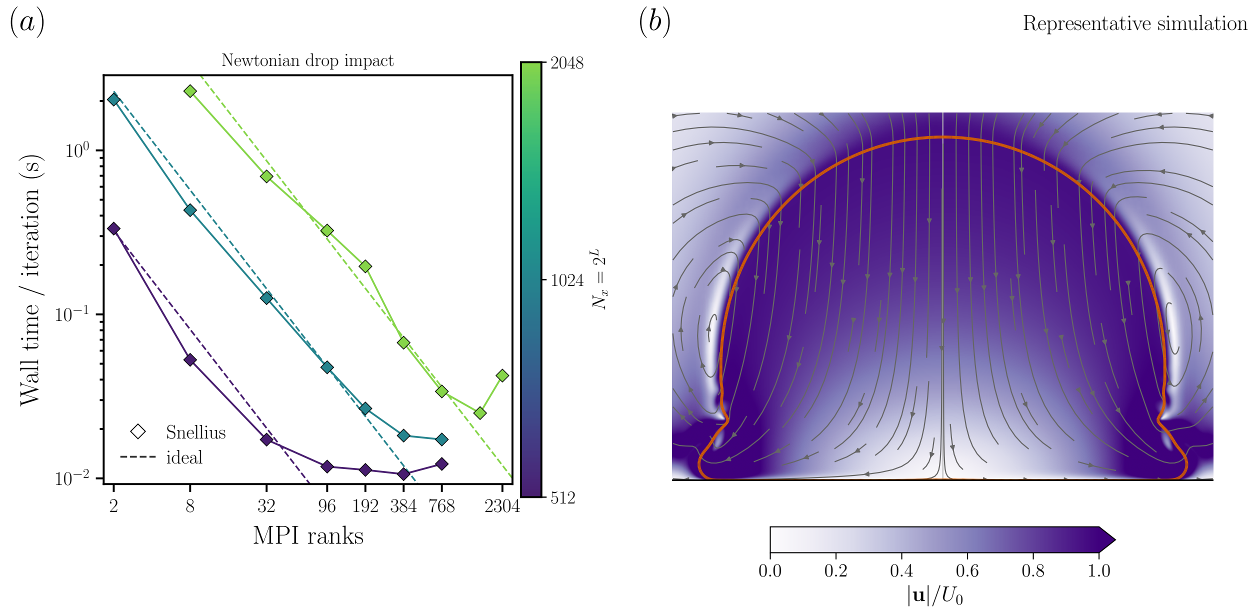


Figure 7: (a) Strong scaling of the uniform-grid Newtonian drop-impact kernel on Snellius Genoa. Axisymmetric VOF with surface tension at $L = 9$, 10 and 11; colour encodes $N_x = 2^L$. Timed window: ten solver steps after geometric initialisation of the drop. The $L = 11$ series starts at eight ranks. (b) Representative impact at $We_D = 200$, $Oh_D = 0.01$ and $tU_0/R = 0.44$, with diameter-based Weber and Ohnesorge numbers. Here $R$ is the initial drop radius and $U_0$ the impact speed. Purple shows speed, grey curves are instantaneous streamlines, orange marks the interface and black marks the substrate. The colour scale is saturated above $U_0$.

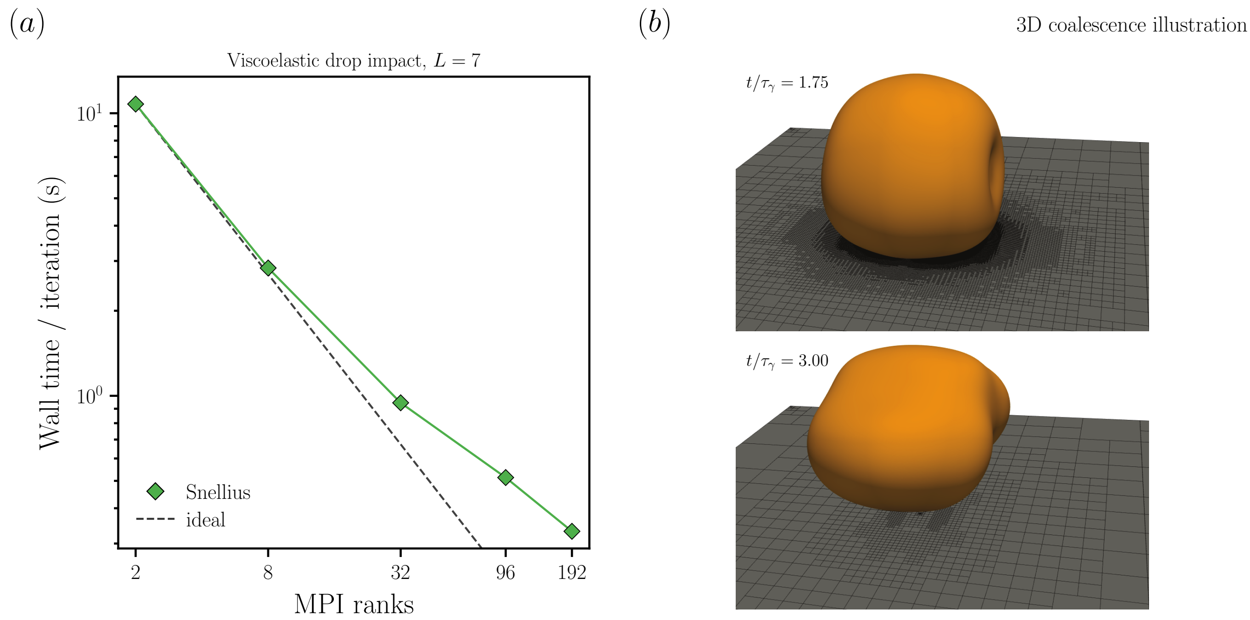


Figure 8: (a) Strong scaling of the uniform-grid three-dimensional viscoelastic drop-impact kernel on Snellius Genoa. Full octree at $L = 7$. Timed window: ten solver steps after geometric initialisation of the drop. (b) Separate illustrative three-dimensional coalescence-induced jumping of two equal drops at $Oh = 0.05$, $Bo = 0$ and $t/\tau_\gamma = 1.75$ and 3.00, where $\tau_\gamma = \sqrt{\rho R^3/\sigma}$. Orange shows the free surface and the substrate plane displays the adaptive mesh. These source images illustrate three-dimensional interface geometry; they are independent of the viscoelastic-impact timing calculation in panel (a).

## 7 Reproducibility and reporting

A procurement-quality result should record the following alongside every timing:

- the exact source revision and unmodified stock-test provenance;
- compiler, optimization flags, MPI implementation and launcher;
- node type, ranks per node, thread affinity and rank binding;
- mesh dimensionality, level, total cell count and cells per rank;
- wall time, MPI time, iteration count and any initialization excluded from the timed region.

The current CSV files contain one observation for each configuration. They are sufficient to locate scaling regimes and design the next campaign, but they do not quantify system variability.

## 8 Conclusion

Basilisk provides a compact CPU benchmark with a direct connection to production multiphase-flow workloads. The stock Poisson and Laplacian kernels separate local computation from MPI communication, while the Marangoni case is compared with reference data and the four uniform-grid cases extend timing coverage across the two-phase solver stack. Existing measurements cover the small class and bracket a 64-node, 192-core-per-node test. The next useful measurement is the VE3D $L = 10$ continuation at 16, 32 and 64 nodes, conditional on the memory check and observed scaling. An $L = 11$ case at 128 nodes can then be considered only if the $L = 10$ problem leaves too little work per rank there; extension beyond 128 nodes remains optional and would require the same evidence. Current timings remain single observations and do not measure run-to-run variability. This distinction keeps present evidence separate from the capabilities that a future system must demonstrate.

## References


[1] S. Popinet, *Basilisk: MPI circle benchmark*, `https://basilisk.fr/src/test/mpi-circle.c`.

[2] S. Popinet, *Basilisk: MPI Laplacian benchmark*, `https://basilisk.fr/src/test/mpi-laplacian.c`.

[3] S. Popinet, An accurate adaptive solver for surface-tension-driven interfacial flows, *Journal of Computational Physics* **228** (2009), 5838–5866. doi:10.1016/j.jcp.2009.04.042.

[4] M. O. Abu-Al-Saud, S. Popinet and H. A. Tchelepi, A conservative and well-balanced surface tension model, *Journal of Computational Physics* **371** (2018), 896–913. doi:10.1016/j.jcp.2018.02.022.